

Frames, Forces, Constraints, and Dissipation

A typed mechanics interface under observation maps

Stas

Independent Research Program; ORCID: 0009-0000-2294-705X

GO Core mechanics interface v0.1 — 28 July 2026

Status. Normative infrastructure for GO Core v0.2. This interface fixes frame direction, point-vector separation, constraint signs, force/power duality, and the nonnegative convention for dissipation. It supplies a common contract for the Foucault, bobsleigh, and roller-coaster modules. It does not propose a new law of mechanics.

Abstract

Mechanical observation models become ambiguous when positions, displacements, forces, generalized forces, and accelerations are added without a common frame or when “dissipated power” is assigned both signs in the same document. We define a typed interface on a configuration manifold Q . A frame edge $A_{F \leftarrow G}(t) \in \text{SO}(3)$ maps free-vector components from G to F , while a point map additionally carries an origin. Differentiating the affine point map yields the translational, Coriolis, Euler, and centrifugal terms; these are representation terms, not new interactions. Physical forces are covectors through virtual power, and generalized forces belong to T_q^*Q . Equality constraints, unilateral contact, and Coulomb-type tangential reactions are given separate signatures. Dissipation is defined by

$$P_{\text{diss}} := -\langle Q_d, \dot{q} \rangle \geq 0,$$

so a quadratic Rayleigh function produces $P_{\text{diss}} = 2\mathcal{R}$. The Lagrangian energy identity is

$$\dot{E}_L = P_{\text{act}} - P_{\text{diss}} + P_{\text{con}} - \partial_t L.$$

For an ideal scleronomous constraint $P_{\text{con}} = 0$; for a moving constraint it need not vanish. The interface also separates physical frame changes, deterministic observation maps, stochastic measurement channels, and estimators. Its statements are standard mechanics recast as a machine-checkable contract.

Contents

1	Scope and typed objects	2
2	Frame graph and affine point maps	2
3	Configuration embeddings and constraints	3
4	Forces, pullback, and virtual power	4
5	Dissipation and contact friction	4
6	Lagrange-d’Alembert energy identity	5
7	Observation protocol and transformation classes	6

1 Scope and typed objects

Let Q be a finite-dimensional configuration manifold and let F be a set of declared Euclidean frames. The interface distinguishes the following objects.

Object	Mathematical type	Admissible operation
Point p	affine-space element	affine frame map, point difference
Free vector v	$V_F \simeq \mathbb{R}^3$	linear frame map, norm, dot product
Configuration velocity	$\dot{q} \in T_q Q$	pairing with a covector
Generalized force	$Q_f \in T_q^* Q$	virtual power $\langle Q_f, \dot{q} \rangle$
Spatial force	$F^F \in V_F$	dot product with a coframed velocity
Acceleration	$a^F \in V_F$	addition only in one declared frame
Observation	map or Markov kernel	may be non-injective and lossy

Boundary 1.1 (Type firewall). A frame transform is an invertible change of representation. An observation map may be many-to-one. Neither is a force law. A generalized force is a covector and cannot be added directly to a spatial acceleration. Multiplication by a mass metric or an explicit pullback is required.

2 Frame graph and affine point maps

Definition 2.1 (Directed frame edge). For frames F, G , the matrix

$$A_{F \leftarrow G}(t) \in \text{SO}(3)$$

maps components of a free vector from G to F : $v^F = A_{F \leftarrow G} v^G$. It obeys

$$A_{F \leftarrow G}^{-1} = A_{G \leftarrow F} = A_{F \leftarrow G}^T, \quad A_{F \leftarrow H} = A_{F \leftarrow G} A_{G \leftarrow H}.$$

A point needs an origin in addition to a rotation. Fix an inertial frame I , let $c_F^I(t)$ be the origin of F represented in I , and abbreviate $A(t) = A_{I \leftarrow F}(t)$. Then

$$x^I = c_F^I + A x^F. \quad (1)$$

The translation is forbidden for a free vector: $v^I = A v^F$, not $c_F^I + A v^F$.

Definition 2.2 (Angular velocity convention). The F -components of angular velocity are defined by

$$A^T \dot{A} = [\Omega^F]_{\times}, \quad [\Omega]_{\times} u = \Omega \times u.$$

Thus $\dot{A} = A [\Omega^F]_{\times}$.

Theorem 2.3 (Kinematics of a moving frame). *If $x^F(t)$ is twice differentiable, then*

$$v^I = \dot{c}_F^I + A(v^F + \Omega^F \times x^F), \quad (2)$$

$$a^I = \ddot{c}_F^I + A(a^F + 2\Omega^F \times v^F + \dot{\Omega}^F \times x^F + \Omega^F \times (\Omega^F \times x^F)). \quad (3)$$

Proof. Differentiate equation (1), substitute $\dot{A} = A[\Omega^F]_\times$, and differentiate once more. The product rule gives two equal cross terms $\Omega^F \times v^F$, producing the coefficient 2. $\square$

For a particle of mass m with physical force $F^F = A^\top F^I$, Newton's equation in F is

$$\begin{aligned} ma^F = F^F - mA^\top \ddot{c}_F^I - 2m\Omega^F \times v^F \\ - m\dot{\Omega}^F \times x^F - m\Omega^F \times (\Omega^F \times x^F). \end{aligned} \quad (4)$$

The last four terms are respectively translational, Coriolis, Euler, and centrifugal effective forces.

Boundary 2.4 (Interpretation). The effective terms in equation (4) are produced by the time-dependent representation. They must not be inserted in the inertial equation and then transformed again.

3 Configuration embeddings and constraints

Let a mechanical realization be a possibly time-dependent embedding

$$\Phi_t : Q \longrightarrow (\mathbb{R}^3)^n, \quad q \longmapsto (x_1(t, q), \dots, x_n(t, q)),$$

with masses $m_a > 0$. For a time-independent embedding, the pullback mass form is

$$G_{ij}(q) = \sum_{a=1}^n m_a \partial_i x_a \cdot \partial_j x_a. \quad (5)$$

Proposition 3.1 (Mass-form rank). *The matrix $G(q)$ in equation (5) is positive semidefinite. It is positive definite exactly when $d\Phi_q : T_q Q \rightarrow (\mathbb{R}^3)^n$ is injective.*

Proof. For $\xi \in T_q Q$,

$$\xi^\top G \xi = \sum_a m_a \|d\Phi_q(\xi)_a\|^2 \geq 0.$$

It vanishes exactly on $\ker d\Phi_q$. $\square$

Equality constraints are written

$$h(t, q) = 0, \quad A_h(t, q) = \partial_q h.$$

For a regular ideal constraint with multiplier λ , the reaction covector is

$$Q_h = A_h^\top \lambda.$$

An admissible velocity satisfies

$$A_h \dot{q} + \partial_t h = 0. \quad (6)$$

Therefore its reaction power is

$$P_h = \langle Q_h, \dot{q} \rangle = -\lambda^\top \partial_t h. \quad (7)$$

It vanishes for a scleronomous constraint $\partial_t h = 0$, but not in general for a moving support.

For unilateral contact we fix the sign convention

$$g_j(t, q) \geq 0, \quad N_j \geq 0, \quad N_j g_j = 0, \quad (8)$$

where $g_j > 0$ means separation and N_j is compressive. Impacts, restitution, and measure-valued impulses require a separate nonsmooth extension and are outside the smooth interface.

4 Forces, pullback, and virtual power

Suppose the spatial velocity of application point a is

$$v_a = J_a(t, q)\dot{q} + v_{a,0}(t, q).$$

A spatial force F_a pulls back to

$$Q_a = J_a^\top F_a \in T_q^*Q.$$

For the virtual part,

$$\langle Q_a, \delta q \rangle = F_a \cdot J_a \delta q.$$

The actual spatial power contains the affine contribution:

$$F_a \cdot v_a = \langle Q_a, \dot{q} \rangle + F_a \cdot v_{a,0}.$$

Consequently a time-dependent embedding cannot be treated as a time-independent Jacobian without losing support power.

Proposition 4.1 (Orthogonal covariance of power). *For $A \in \text{SO}(3)$, if $F^I = AF^F$ and $v^I = Av^F$, then*

$$F^I \cdot v^I = F^F \cdot v^F.$$

Proof. $(AF^F)^\top (Av^F) = (F^F)^\top A^\top Av^F$. □

This proposition applies to coframed physical vectors. It does not authorize pairing a force in one frame with a velocity in another.

5 Dissipation and contact friction

Definition 5.1 (Dissipative generalized force). A generalized force $Q_d(t, q, \dot{q})$ is dissipative on an admissible domain if

$$\langle Q_d, \dot{q} \rangle \leq 0.$$

Its nonnegative dissipated power is

$$P_{\text{diss}} := -\langle Q_d, \dot{q} \rangle \geq 0. \quad (9)$$

For a Rayleigh function

$$\mathcal{R}(t, q, \dot{q}) = \frac{1}{2} \dot{q}^\top C(t, q) \dot{q}, \quad C = C^\top \succeq 0,$$

the convention is

$$Q_d = -\partial_{\dot{q}} \mathcal{R} = -C\dot{q}, \quad P_{\text{diss}} = \dot{q}^\top C\dot{q} = 2\mathcal{R}.$$

At contact j , let n_j be the unit normal and $P_{T,j} = I - n_j n_j^\top$. Write the force on the modeled body as

$$F_j = N_j n_j + T_j, \quad T_j \cdot n_j = 0,$$

and the tangential relative velocity as

$$v_{T,j} = P_{T,j}(v_{\text{body},j} - v_{\text{support},j}).$$

A Coulomb-type admissible reaction obeys

$$\|T_j\| \leq \mu_j N_j, \quad T_j \cdot v_{T,j} \leq 0. \quad (10)$$

Thus

$$P_{\text{fric},j} = -T_j \cdot v_{T,j} \geq 0. \quad (11)$$

For sliding with $v_{T,j} \neq 0$, the maximum-dissipation selection is

$$T_j = -\mu_j N_j \frac{v_{T,j}}{\|v_{T,j}\|}.$$

For sticking, $v_{T,j} = 0$, the friction force is set-valued inside the cone and its instantaneous power is zero.

Definition 5.2 (Guarded cone utilization). On an admissible contact state define

$$u_j = \begin{cases} \|T_j\| / (\mu_j N_j), & \mu_j N_j > 0, \\ 0, & \mu_j N_j = 0 \text{ and } T_j = 0. \end{cases}$$

The cone law excludes the missing case $\mu_j N_j = 0$, $T_j \neq 0$. For a finite contact set,

$$S_{\text{slip}} = 1 - \max_j u_j \in [0, 1],$$

with $\max \emptyset := 0$.

6 Lagrange-d'Alembert energy identity

Let $L(t, q, \dot{q})$ be regular on the active smooth regime and set

$$E_L = \langle \partial_{\dot{q}} L, \dot{q} \rangle - L.$$

The forced equations are

$$\frac{d}{dt} \partial_{\dot{q}} L - \partial_q L = Q_{\text{act}} + Q_d + Q_{\text{con}}. \quad (12)$$

Theorem 6.1 (Energy balance). *Every C^2 solution of equation (12) satisfies*

$$\dot{E}_L = P_{\text{act}} - P_{\text{diss}} + P_{\text{con}} - \partial_t L, \quad (13)$$

where

$$P_{\text{act}} = \langle Q_{\text{act}}, \dot{q} \rangle, \quad P_{\text{con}} = \langle Q_{\text{con}}, \dot{q} \rangle.$$

Proof. Direct differentiation gives

$$\dot{E}_L = \left\langle \frac{d}{dt} \partial_{\dot{q}} L - \partial_q L, \dot{q} \right\rangle - \partial_t L.$$

Insert equation (12) and use equation (9). □

Corollary 6.2 (Autonomous ideal case). *If $\partial_t L = 0$, $Q_{\text{act}} = 0$, $Q_d = 0$, and the constraints are ideal and scleronomous, then $\dot{E}_L = 0$.*

Boundary 6.3 (No-double-counting rule). A moving track may be represented either by an ambient formulation with explicit reaction/support power or by reduction to a time-dependent Lagrangian. The same support work must not be added in both places.

7 Observation protocol and transformation classes

A complete protocol is the typed chain

$$X_t \xrightarrow{\mathcal{R}_{\text{red}}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

The protocol record λ includes:

- (i) frame graph, clock, coordinates, and unit context;
- (ii) system descriptor and constraint convention;
- (iii) spatial and temporal resolution and observation horizon;
- (iv) deterministic readout and stochastic/noiseless declaration;
- (v) calibration, estimator, uncertainty, and missing-data policy;
- (vi) a declared transformation group for every invariant claim.

Boundary 7.1 (Invariant scope). Rigid-motion invariance, covariance under a time-dependent frame, Galilean invariance, label-swap invariance, and invariance under a lossy observation map are different statements. The word “invariant” is rejected unless the acting group and the transformed objects are named.

8 Machine gates

The mechanics extension admits a reference document only if the following gates are satisfied.

Gate	Required declaration	Rejected pattern
Frame closure	every vector sum has one frame; every point map has an origin	$r^I + q^B$ without $A_{I \leftarrow B}$
Force type	spatial force or generalized covector, with pullback map	adding F directly to a
Constraint sign	gap side, reaction sign, complementarity, active set	unsigned g, N convention
Dissipation sign	$P_{\text{diss}} \geq 0$ and force-on-system convention	$\dot{E} = -\sum T \cdot v_{\text{rel}}$ with $T \cdot v_{\text{rel}} \leq 0$

Gate	Required declaration	Rejected pattern
Moving support	explicit $\partial_t h$, support power, or reduced time-dependent L	declaring every ideal constraint workless
Friction guard	zero-load branch for $\ T\ /(\mu N)$	division by $\mu N = 0$
Observation split	frame map, observation, channel, estimator kept separate	calling a camera projection a frame transform
Invariant scope	named group and hypotheses	unqualified “observation invariant”

9 Claim boundary

The interface is a consistency layer. It does not establish:

- existence or uniqueness for arbitrary nonsmooth contact;
- a constitutive law for ice, wheel-rail, or air resistance;
- structural safety, fatigue, or human-tolerance limits;
- a precision geophysical pendulum model;
- scientific novelty of the underlying standard identities.

Any application must add its own regularity, constitutive, parameter, and uncertainty hypotheses.

References

- [1] A. M. Bloch, *Nonholonomic Mechanics and Control*, 2nd ed., Springer, 2015. doi:10.1007/978-1-4939-3017-3.
- [2] B. Brogliato, *Nonsmooth Mechanics: Models, Dynamics and Control*, 3rd ed., Springer, 2016. doi:10.1007/978-3-319-28664-8.
- [3] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed., Springer, 1999. doi:10.1007/978-0-387-21792-5.
- [4] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Springer, 1989. doi:10.1007/978-1-4757-2063-1.